

Prevent Data Loss by Using FOSS Tools



Vital data can be lost because of some physical damage to the storage hardware or due to a virus. That's why it should be stored using a suitable backup method and on a reliable storage medium. There are a number of open source software tools to help businesses store data in an automated way. A few of these are discussed in this article.

Data can be backed up in various ways, but the most commonly used types of backup are full backup, differential backup, incremental backup and reverse incremental backup. Most of these backups can be done manually or be automated using backup software, and kept in the repository. A repository is a place where data is kept safely on storage media in an organised manner.

Full backup

Here, every time a backup is taken, all the stored data is copied from the source storage (usually the hard drive) to the storage media, and kept in the repository. For example, if on Day 1, a full backup is done for all the stored data, then on Day 2 also, all the stored plus changed data will be copied to the backup storage media and kept in the repository. The same will be done on Day 3, Day 4, and so on. This type of backup is reliable but needs a lot of storage space on a regular basis.

Differential backup

A differential backup is done by comparing the differences in data between the first full backup and the changed source data. At first, a full backup is created, and then differential backups are created. Only the data that changed since the *last full backup* is copied to the storage media. So

every time the data gets changed on the source storage, a differential backup of only the changed data is created and stored in the repository.

For example, on Day 1, a full backup is created for the source data. If there is any change in this data (compared to the last full backup) then another backup for only the changed data is created on Day 2. On Day 3, a backup of all the data that changed since Day 1 is taken, and this is repeated on Day 4, and so on. This type of backup is faster than a full backup and initially takes less storage space, but the backup file may grow bigger in size after a certain number of backups. That's why this type of backup is usually done on a HDD or a tape.

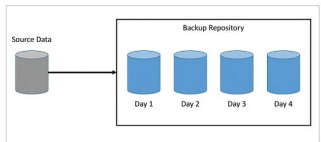


Figure 1: Full backup

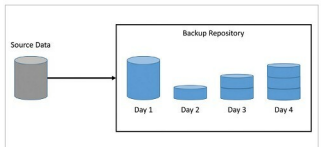


Figure 2: Differential backup